ST-Buddy: Designing and Evaluating a Course-Grounded LLM Chatbot for Academic and Administrative Support

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Abstract

Understaffed universities and limited funding for tutors risk students' disengagement due to missing assistance. HCI research increasingly focuses on conversational agents for scalable educational support. This paper presents a course-specific AI chatbot designed to answer students' questions and reduce tutor workload by addressing frequent organizational and content-related questions. Developed as a pilot case study for an introductory Software Technology course, *ST-Buddy* combines modular LLM-based dialogue with a flexible course-grounded knowledge integration to enable adaptive, context-aware responses grounded in relevant course material and logistical information. A formative evaluation of *ST-Buddy* (n=32) showed good usability (SUS=77.81), perceived helpfulness particularly when tutors were unavailable, and rated response quality as relevant and understandable. Additionally, students' self-reported everyday academic AI use and lessons learned from the evaluation settings are discussed. These findings outline implications for improving future evaluations and highlight the potential of modular chatbot frameworks to scale personalized support in large courses.

CCS Concepts

• **General and reference** → Empirical studies; • **Applied computing** → **E-learning**; • **Human-centered computing** → *Empirical studies in HCI*; *Text input*; *Field studies.*

Keywords

User Expectations, Chatbot Design, AI-based Chatbot, AI-based Support Agent, Conversational Agents, Large Language Models, Course-Specific AI Support, Generative AI, User-Centered Design, Immediate Feedback, In-class Evaluation, Online Evaluation, Educational Technology, Participatory Design, Computer Science Education

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1 Introduction

Across higher education, many students require timely and individualized support to navigate academic and organizational demands, particularly during the early stages of their studies and in large introductory courses with heterogeneous cohorts [14, 30]. This need is especially pronounced for specific student groups with diverse backgrounds and expectations [5]. While study success is closely linked to students' ability to regulate their learning processes and access relevant information when needed [23, 32], institutional support structures often struggle to meet this demand at scale. As a result, students may experience uncertainty, delayed feedback, and fragmented access to information, which can negatively affect engagement and persistence [14]. Peer learning, tutorials, and office hours are widely regarded as effective, diversity-sensitive, and inclusive forms of support [25]. However, they are inherently resource-bound and frequently constrained by staffing shortages, financial limitations, and increasing student numbers. These limitations have motivated Human-Computer Interaction (HCI) researchers and educators to increasingly explore Conversational

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Agents (CAs) as scalable tools for on-demand assistance in educational contexts [1, 30]. Recent research further highlights the potential of Artificial Intelligence (AI)-based tools to foster inclusion [19], better support students in large courses [18], promote problem-solving skills [27], improve formative feedback processes [31], or enable conversational, adaptive, and personalized interactions [21]. However, deploying such systems in higher education introduces new challenges related to interaction design, trust, pedagogical alignment, and data privacy. Without careful user-centered design and transparent implementation, AI-based support risks remaining generic, opaque, or misaligned with students or course-specific needs [5].

This paper addresses these challenges through the design and evaluation of a course-specific Large Language Model (LLM)-based chatbot, called *ST-Buddy*. *ST-Buddy* is explicitly not designed as an intelligent tutor or coach, but as a course-specific support agent complementing existing human assistance. More specifically, *ST-Buddy* provides scalable learning and organizational support to complement human assistance by offloading recurring questions, and is employed in an introductory Software Technology (ST) course.

Addressing limited design and implementation reporting in prior educational chatbot research [4], *ST-Buddy*'s technical architecture is documented transparently and released as open source.[1]

This research provides contextual insights into students self-reported everyday AI use and addresses the following research question (RQ): How do students perceive and evaluate the course-specific chatbot *ST-Buddy* as support for learning and organizational questions with regard to usability, usefulness, and response quality? To answer this RQ, the paper presents (1) the learner-centered design of *ST-Buddy* and its underlying system decisions (modular bot routing, privacy-aware deployment, and course-aligned Retrieval Augmented Generation (RAG)); (2) findings from a formative in-class evaluation on usability and perceived usefulness; and (3) practical lessons learned for designing and evaluating course-specific educational chatbots in authentic learning contexts.

2 Background and Related Work

2.1 Conversational Agents in Education

Educational chatbots have been studied for tutoring, practice, motivation, and administrative support. Wollny et al. [30] synthesized 74 studies and distinguished three pedagogical roles: (i) *learning-oriented* tutoring and content practice (e.g., language learning [3, 12] and learn about laws [8]), (ii) *assisting* course-related and organizational support [26], and (iii) *mentoring* systems supporting scaffolding and Self-Regulated Learning (SRL) [13]. While results are generally positive, the review highlights persistent limitations, such as weak integration into teaching practice and limited evidence from realistic deployments [30]. Recent work further illustrates the potential of LLM-based tutors: Wang et al. co-designed a clinical reasoning tutor powered by *Med-PaLM 2*[2] and evaluated it with medical learners (N=8), showing that interactive AI tutoring can support practice and feedback when it emulates effective instructor traits such as scaffolding and reflection [28].

2.2 Course-Specific Chatbots and Support for Computer Science Education

Course-specific assistants are particularly relevant for introductory computer science education, where learners need timely explanations, structured guidance, and support when human tutors are unavailable [16, 18]. A recent scoping review identified that most CAs for novice programmers focus on *primary learning support*, and only a few extend to broader student needs [4]. The reviewed systems mainly address (i) *learning-oriented tutoring and practice* (e.g., concept explanations and quizzes [22] or scaffolding-based field studies [15]), (ii) *task-level programming support* such as debugging assistance or feedback in code review activities [10, 20], and only rarely (iii) *secondary support beyond instruction* (e.g., holistic academic/personal support [16] or scalable course-integrated AI support in large cohorts [18]). Across studies, evaluations often rely on brief pilot deployment and short-term user reports rather than longitudinal study design with larger and more diverse samples, and Barzanji et al. further highlight limited reporting of system architectures and design decisions, hindering reuse and comparison [4].

Despite growing work on course-specific and RAG-based chatbots, many systems lack transparent architectural reporting, rely on externally hosted LLM services, give limited attention to institutional privacy requirements, and focus mainly on learning tasks. ST-Buddy addresses these gaps through a scalable and privacy-aware deployment using foundation models hosted within the university infrastructure. The system is released as open source to support reproducibility and reuse, and supports multiple course-related knowledge domains (e.g., different academic contents and administrative information) through topic-aware routing and retrieval, which has not yet been explicitly evaluated in the literature.

3 System Design

ST-Buddy is situated on a self-developed chatbot platform to easily provide CAs tailored to individual contexts - in this case, the introductory Software Technology (ST) course. The platform hosts the CA on a homogenized website developed through user-centered processes and with attention to accessibility and data-privacy. User data and chat messages are solely saved client-side in the browser's local storage, and LLM providers with no-log and no-training policies are selected. The backend is built using a combination of FastAPI[3] for routing, LangChain [7] for structuring LLM calls, LiteLLM[4] for managing available models and routing LLM requests to providers, and LangFuse [24] to trace LLM generations and keep a version history of prompts.

ST-Buddy provides students with access to both course content and organizational information via natural language interaction in German and English, serving both academic and administrative needs. Course-specific materials are embedded and saved in a pgvector[5] vector database alongside the full texts and metadata. Through a RAG process, these materials are retrieved based on the user request. Course data is split into three different categories: content consisting of reading material accompanying the lecture,

[1] ST-Buddy repository: https://github.com/KI-FEM/metis/tree/st-buddy-2507
[2] Med-PaLM 2, https://sites.research.google/gr/med-palm/, accessed on January 16th 2026
[3] FastAPI, https://fastapi.tiangolo.com/, accessed on January 16th 2026
[4] LiteLLM, https://www.litellm.ai, accessed on January 16th 2026
[5] pgvector, https://github.com/pgvector/pgvector, accessed on January 16th 2026

lecture exercises and parts of their solution, and organizational content like FAQs, exam office information, and anonymized forum entries from previous semesters. ST-Buddy dynamically chooses which category of data to load for RAG based on the user input using a pre-generation LLM call.

To increase document retrieval performance in the RAG process, organizational content uses a similarity search, retrieving three documents per query, while lecture content and exercises use a similarity search filtered by a relevance score threshold to further trim results to the most relevant documents. Additionally, user queries are rewritten into four additional queries prior to use in the RAG pipeline through a separate LLM call, as colloquial user inputs may result in inaccurately retrieved documents. Figure 1 shows an overview of the proposed system design used in the following study.

4 ST-Buddy Evaluation

4.1 Study Design

To examine students' everyday use of AI and their perceptions of *ST-Buddy*, a formative mixed-methods evaluation was conducted with students from the target group in two settings: (i) An in-class semi-controlled test during exercise sessions to capture initial reactions and usability issues, and (ii) a four-week online field test, where students used *ST-Buddy* independently for exam preparation and exercise support. Both settings followed the same procedure, consisting of a pre-survey, *ST-Buddy* testing phase, and a post-survey.

Participants. The study involved students enrolled in the introductory ST course at Dresden University of Technology across different study programs. Participation in the study was voluntary and independent of course requirements, grading, or academic evaluation. The chatbot was provided as an optional support tool and was not part of mandatory course activities or assessment. All participants provided informed consent and signed a privacy policy in line with data protection regulations. Survey data were collected anonymously and were not accessible to course instructors at the individual level.

In-Class Testing Procedure. The in-class evaluation was conducted as a semi-controlled test of the implemented CA-based prototype. After a short introduction, participants accessed the chatbot website and completed a pre-survey hosted on a separate online survey platform, capturing demographics and prior AI use, measuring the standardized Affinity for Technology Interaction (ATI) scale [11]. Survey links contained an embedded anonymous ID that was automatically pre-filled, enabling matching of responses across surveys. After submitting the pre-survey, participants freely interacted with *ST-Buddy*. To lower the entry barrier, course-related conversation-starter questions were provided (e.g., "How can the *State Pattern* in the *BankAccount* exercise be implemented?" and "I am sick. How can I withdraw from the exam?"). Finally, participants completed a post-survey including System Usability Scale (SUS) [6] to assess perceived system usability and Partner Model Questionnaire (PMQ) [9] to evaluate the CA as a conversation partner. Participants also rated whether the chatbot would be useful when no tutor is available, and evaluated response quality (relevance, correctness, clarity, complexity, length, and linguistic naturalness) on 5-point Likert scales from "not at all" to "completely". To complement quantitative measures with qualitative input, participants could optionally provide open-ended feedback on what they liked and disliked about the system.

Online Field Test Procedure. The online field test followed the same survey design and instruments. Participants received the chatbot link and project information via the course newsletter. To support self-directed use, the pre-survey remained accessible during the first two weeks of the test period, while the post-survey was activated only at the end of the study and distributed via a follow-up newsletter.

4.2 Results

4.2.1 Participation. A total of 103 participants started the pre-survey, including 45 in the semi-controlled in-class evaluation and 60 in the online field test. In the in-class setting, 43 participants completed the pre-survey; 38 started the post-survey, and 32 completed it. Of the 32 participants, about 20% (n=6) were female and about 80% (n=26) male. The majority (65.6%; n=21) were between 18 and 20 years old, while the remaining 11 (34.4%) were between 21 and 29 years old. Most students (68.75%; n=22) were enrolled in *(Media) Computer Science*, with the rest being split between *Business Information Systems*, *Engineering*, and *Mathematics*. In the online field test, which was sent to over 300 course members, 22 participants completed the pre-survey, 12 started the post-survey, and 2 completed it, of which one contained invalid responses. Since the two study setups were different and the online field test had a low response rate, it was excluded from analysis, and the results reported below are based on the 32 valid responses from the in-class evaluation.

4.2.2 Pre-survey Results. The ATI showed a mean of $M = 4.32$ ($SD = 0.62$). Regarding AI use, 50% of participants reported using AI daily or almost daily, 28.1% several times per week, 15.6% occasionally, and 6.3% rarely. In students' open-ended reflections on their previous experiences with AI tools, participants were mostly positive about their experiences with AI use, though the issues of hallucinations, especially in complex tasks, were noted. Participants reported different purposes of using AI, although all participants (96.9%; n=31) indicated that they use AI to explain concepts or tasks, followed by coding/programming support (62.5%; n=20) and exam preparation (53.1%; n=17). Summarization of lecture content was less common (31.3%; n=10). In addition, some participants (15.6%; n=5) mentioned "other" uses such as translation, text correction, image generation, and information search.

4.2.3 Post-survey Results. Regarding response quality, 65.6% (n=21) of participants selected "mostly" or "completely" for relevancy and clarity, and 71.9% (n=23) estimated the answers as completely or mostly correct. Regarding appropriateness of length, 62.5% (n=20) answered "mostly" or "completely", with 21.9% (n=7) selecting "mostly not" or "not at all". Ratings of linguistic naturalness showed that 62.5% (n=20) answered "mostly" or "completely". Perceptions of complexity were balanced, with 34.4% (n=11) rating responses "not at all" or "mostly not", 31.3% (n=10) as "partly", and 34.4% (n=11) as

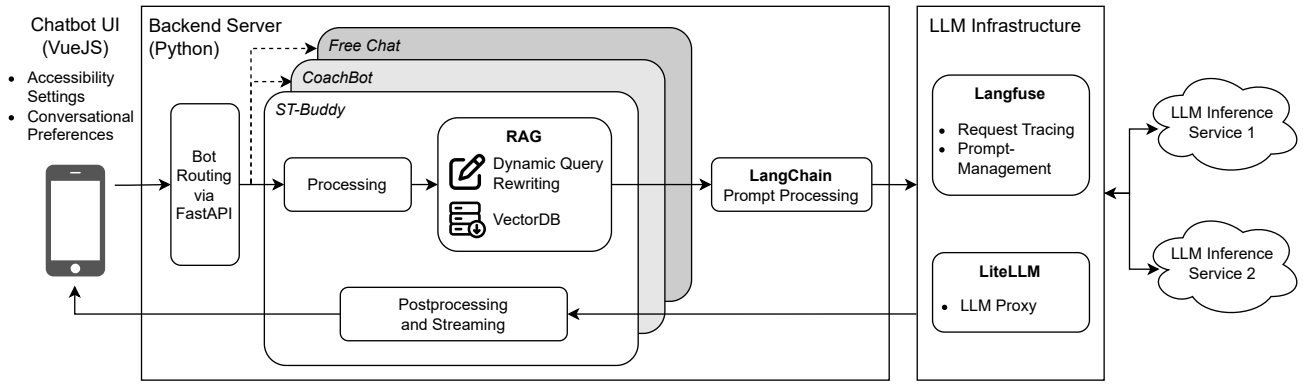


Figure 1: ST-Buddy's Technical Infrastructure

"mostly" or "completely". 75% of students evaluated the CA's usefulness when no tutors are available positively (n=17 for "mostly", n=7 for "completely"), 15.6% (n=5) remained neutral, and 9.4% (n=3) selected "not at all" or "mostly not".

Usability, measured with the SUS, revealed a score of $M = 77.81$ ($SD = 12.47$), which can be labeled as "Good" [2]. The mean scores across the three subscales of the PMQ were 5.07 for communicative competence, 4.06 for human-likeness communication, and 4.08 for communicative flexibility. The overall PMQ score averaged 4.57 on the 7-point scale.

The open-ended responses on what students liked and disliked about *ST-Buddy* were analyzed using thematic coding, resulting in nine themes. The most frequent theme was *relevance and course-specificity* (n=11), with students emphasizing that the chatbot was grounded in lecture material. Further prominent themes concerned positive *answer quality and clarity* (n=9) and *answer accuracy and completeness* (n=9). Students also valued the *efficiency and ease of use* (n=8) and *practical examples* (n=6) that ST-Buddy gave. On the limitations side, participants highlighted *technical issues* (n=10), such as duplicated or aborted responses, and issues with *response length and style* (n=7), e.g. long and complex answers. Furthermore, *missing features and usability gaps* (n=6), such as an "option to abort answers" (P29), and *lack of visual support* (n=5) were identified, explicitly requesting diagrams for programming contexts.

5 Discussion

5.1 How Students Self-report Current AI Use

Pre-survey results indicate that AI tools are embedded in students' study routines, with most participants reporting frequent (often daily) use, primarily for explanations, programming support, and exam preparation, while summarization was less common. This aligns with broader trends in higher education [15, 18], and reflects the *learning* and *assisting* roles of educational chatbots described by Wollny et al. [30], but also reflects ambivalence between efficiency benefits and accuracy concerns [29], indicating the need for AI literacy and transparent support tools.

5.2 Course-Specific Value and Design Requirements

Given students' frequent use of AI for academic tasks, the positive evaluation of *ST-Buddy* suggests that course-specific AI assistance for internal learning and organizational needs, which general-purpose chatbots typically do not cover, is valued, aligning with prior research on scalable support in large courses [5, 18]. Given the prototype status and short exposure time, usability was rated positively (SUS=77.8), and 75% of students perceived *ST-Buddy* as a useful alternative when tutors are unavailable. However, this result may have been influenced by the cohort's high technology affinity (ATI=4.32) and suggests low interaction barriers rather than long-term acceptance. Student feedback further highlighted key HCI requirements for sustained use, including support beyond text-only output and robust mobile usability, as mobile learning is increasingly common [17].

5.3 Study Design Insights - Implications for Empirical Research

While the online field test was intended to complement the in-class study, its primary contribution lies in methodological insights rather than quantitative results. The strong contrast between high in-class participation and low online completion likely reflects context, technical constraints, and study design. Open-ended feedback indicated that technical issues occurred for 6/32 participants, and online participation may have been further hindered by access restrictions (university network/VPN). Conducting the field test prior to and during the exam period and using a multi-stage survey setup with lacking incentives may have additionally reduced engagement and increased dropout. This underscores the importance of timing and low-friction designs when evaluating educational CAs, and motivates combining semi-controlled formative studies with longitudinal field evaluations.

5.4 Limitations

As a first-iteration prototype, *ST-Buddy* occasionally produced technical issues (e.g., duplicated or incomplete responses), which may have affected perceived reliability. Generalizability is limited due to participants predominantly being male, German, and computer

science students. Moreover, the privacy-focused local-first design led to mismatched pre-/post-survey data when participants used multiple devices. Although the online study invitation reached over 300 students, only one participant completed both surveys; therefore, comparisons between online and in-class settings were not feasible. Future studies should include a post-study step, asking online participants about reasons for dropout to inform improvements in study design and recruitment. Overall, the results rely on the in-class sample ($n = 32$), limiting subgroup analyses. Due to the short (30-minute) test phase and the absence of a control group, conclusions about learning efficacy and causal effects are restricted. A study setup testing ST-Buddy against a group using a general-purpose AI chatbot would cast a light on the efficacy of course-specific knowledge integration. Future work should involve larger and more diverse samples, include control groups, and integrate objective learning measures. While the evaluation focused on perceived usability and usefulness, future work should also empirically compare architectural variants (e.g., RAG vs. non-RAG) to isolate design effects.

6 Conclusion

This paper presented *ST-Buddy*, a course-specific AI-based chatbot designed to provide grounded learning and organizational support in a large introductory Software Technology course, addressing students' recurring struggle to identify relevant and reliable course information at the moment of need. The reusable system design of *ST-Buddy* builds on modular LLM-based dialogue with contextually relevant information injected using RAG and is available as open-source software. A formative in-class evaluation with 32 participants showed good perceived usability (SUS=77.8), positive usefulness when tutors are unavailable, and largely favorable ratings of response relevance, clarity, and correctness. Beyond system design and performance, insights into students' everyday AI use and expectations, and valuable lessons learned for in-class versus field-test evaluations with student target groups are discussed.

Taken together, combining grounded AI architectures with HCI-based evaluation and student feedback is critical for moving toward robust, course-grounded educational support systems in real-world teaching contexts.

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